Diffusion-Based Handwritten Image Deblurring and Restoration

Using U-Net and Perceptual Loss for English and Bangla Handwriting

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***Abstract*—Blurred handwritten documents are difficult to read and are less suitable for digitization and optical character recognition (OCR). This work presents a U-Net-based framework for deblurring English and Bangla handwritten document images. English training pairs are formed from the HCCD dataset using its three blur levels, while Gaussian blur with a kernel size of 15 is applied to clear Bangla samples to create synthetic pairs. The combined dataset contains 1,898 blurred–clean pairs. Images are resized to 512 $\times$ 512 pixels, and a 7.7-million-parameter U-Net is trained for 12 epochs on an NVIDIA T4 GPU using Adam with a weighted combination of L1, structural similarity (SSIM), and VGG perceptual losses. The best validation loss of the combined model is 0.0477, and an L1-only baseline on the HCCD validation split reaches about 35.89 dB PSNR with an SSIM above 0.96. A preliminary conditional diffusion experiment was also explored but was not carried into the final system. The trained model is deployed through a Gradio web interface for image upload and restoration. This paper presents the dataset preparation, model, training procedure, observed results, limitations, and possible improvements.**

***Index Terms*—Image deblurring, handwritten document restoration, U-Net, SSIM, VGG perceptual loss, HCCD, Bangla handwriting, Gradio.**

I. Introduction

Handwritten documents remain important in education, administration, personal record keeping, and historical archives. Their usefulness can be reduced when the writing is captured with poor lighting, an out-of-focus camera, or a low-quality scanner. Such acquisition conditions can produce blurred strokes and reduce the readability of individual characters. The problem is particularly relevant when documents are digitized using ordinary mobile cameras rather than dedicated scanning equipment.

Image deblurring aims to recover a sharper image from a degraded observation. In handwritten documents, restoration must preserve thin character strokes, spaces, punctuation, and the local structure of the writing. Excessive smoothing may remove useful details, while aggressive sharpening may introduce artificial edges. Therefore, a restoration model should consider both pixel-level similarity and perceptual structure.

This project develops a U-Net-based deblurring approach for English and Bangla handwriting. The English portion uses the HCCD dataset and its available blur levels. The Bangla portion contains clear handwriting images for which synthetic Gaussian blur is generated to create paired training data. A single model is trained on the combined collection and is made available through a Gradio interface, where a user can submit an image and view the restored output.

A. Project Scope

The work focuses on paired image restoration at a fixed input resolution of 512 $\times$ 512 pixels. It covers dataset pairing, preprocessing, U-Net training, a combined loss function, validation, visual inspection, and web-based inference. The Bangla training examples use synthetic Gaussian blur; they should not be interpreted as a collection of naturally blurred Bangla photographs.

B. Objectives

The main objectives are:

- prepare paired blurred and clean handwriting images for English and Bangla;
- train a single U-Net restoration model for both language datasets;
- combine pixel, structural, and perceptual losses during optimization;

- evaluate the work using validation loss, PSNR, SSIM, and visual examples; and
- make the trained model accessible through a Gradio interface.

II. Problem Definition and Proposed Solution

A. Problem Definition

A blurred handwritten image may contain overlapping or weakened character strokes. This makes manual reading difficult and can affect later OCR processing. Supervised image restoration requires pairs in which each degraded input has a corresponding clean target. Such paired examples are not always available for low-resource languages, including Bangla. The project therefore uses different pairing strategies for the two language sources.

B. Proposed Solution

For English handwriting, the available HCCD blur levels (1, 2, and 3) are matched with their corresponding ground-truth images. For Bangla handwriting, Gaussian blur is applied to clear source images to construct synthetic blurred–clean pairs. Both sources are combined and passed through the same preprocessing pipeline. A feed-forward U-Net recovers a clean image from each blurred input in a single forward pass. During training, the prediction is compared with its clean target using L1, SSIM, and VGG perceptual losses.

C. Main Components

The implementation consists of five practical components: (1) image collection and pairing, (2) resizing, (3) restoration using a U-Net, (4) loss calculation and parameter optimization, and (5) inference through a Gradio interface. The complete sequence is illustrated in Fig. 3.

III. Related Work

Early image-deblurring approaches often used mathematical degradation models and restoration filters, including Wiener filtering. Such approaches can be sensitive to assumptions about the blur kernel and may not adequately handle complex image degradation. Learning-based methods instead learn a mapping from degraded images to restored images using examples. Deep learning models such as SRCNN, DeblurGAN, and U-Net have been considered for image restoration. Conditional diffusion models have also been studied for this purpose; they restore an image through iterative denoising, which is more costly at inference time than a single forward pass. U-Net is particularly useful for restoration because its skip connections preserve spatial details that may be lost during down-sampling. SSIM and VGG perceptual losses can also produce sharper and more natural-looking images than standard pixel-wise MSE.

Ronneberger et al. introduced U-Net for biomedical image segmentation, using an encoder–decoder structure and skip connections to retain spatial information [1]. Although the original task was segmentation, the same structural idea is useful in image-to-image restoration: the encoder learns contextual features, while skip connections provide the decoder with details from earlier feature maps.

Structural similarity (SSIM) compares local image structure rather than relying only on absolute pixel differences. Wang et al. introduced SSIM as an image-quality assessment measure [2]. Perceptual losses compare feature representations extracted by a pretrained network. Johnson et al. demonstrated the use of feature-space losses for image transformation tasks [3]. In this project, L1, SSIM, and VGG perceptual loss are combined to encourage pixel accuracy, structural consistency, and perceptual sharpness.

TABLE I
Dataset Composition Used in the Project

Dataset Portion	Source Images	Pairs
English (HCCD)	574 clean images	1,722
Bangla	176 clear images	176
Combined	–	1,898

IV. Methodology

The methodology uses two handwriting sources, which are paired, preprocessed, and combined into a single training collection. The composition of the resulting dataset is summarized in Table I.

A. Dataset Preparation

Two handwriting sources are used. The first is the HCCD English handwriting dataset [4], which contains 574 ground-truth images and three corresponding blur levels. The second source is the SAIPL Bangla handwriting collection [5], from which 176 clear images are used.

English pairing: The three HCCD blur levels are matched to their clean ground-truth images using the image number in each file name, extracted with a regular expression. This produces $574 \times 3 = 1{,}722$ English pairs.

Bangla pairing: The clear Bangla images are retained as clean targets. A Gaussian blur with kernel size 15×15 is applied to each image using OpenCV, which derives the standard deviation automatically from the kernel size ($\sigma \approx 2.6$). This creates 176 synthetic pairs.

Combined collection: The English and Bangla pairs are combined, giving a total of 1,898 pairs. Fig. 1 indicates that the English HCCD portion forms the larger share of the combined dataset. Consequently, the combined training set is not balanced by language.

B. Image Preprocessing

All input and target images are resized directly to 512×512 pixels and converted to three-channel RGB tensors with pixel values in the range $[0, 1]$. The same size and scaling are applied to both language portions so that they can be processed by one model. The resizing does not preserve the original aspect ratio; the restored output is resized back to the original dimensions at inference time.

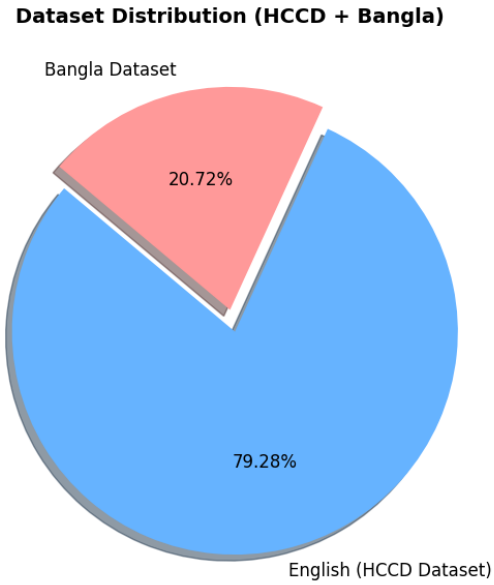


Fig. 1. Distribution of English HCCD and Bangla samples in the combined dataset.

The paired structure is important: the blurred image is used as the network input, while the corresponding clear image acts as the ground-truth target. During training, the model output is compared with this target to calculate the loss.

C. U-Net Model Architecture

The restoration network is a U-Net with an encoder–decoder design. The encoder contains three stages, each built from a DoubleConv block (two 3×3 convolutions, each followed by batch normalization and ReLU) with 64, 128, and 256 channels, followed by 2×2 max pooling. A bottleneck DoubleConv block with 512 channels processes the lowest-resolution representation. The decoder contains three stages; each upsamples with a 2×2 transposed convolution, concatenates the corresponding encoder feature map, and applies a DoubleConv block. A final 1×1 convolution produces three output channels, followed by a sigmoid activation that keeps the output in $[0, 1]$.

Skip connections transfer feature maps from encoder stages to corresponding decoder stages. These connections help preserve spatial details that may otherwise be lost during downsampling. This is important for handwriting because character identity can depend on small stroke differences. The model contains approximately 7.7 million trainable parameters and accepts a three-channel 512×512 image.

Preliminary diffusion experiment: Before the final design was chosen, a conditional denoising diffusion model was also explored. It used a diffusers `UNet2DModel` that received the noisy clean image concatenated with the blurred image (six input channels), a 1000-step noise schedule, and an MSE noise-prediction loss on 128×128 patches of the HCCD data. This experiment was not carried into the final system. All results reported in this paper refer to the feed-forward U-Net described above.

D. Training Objective and Loss Function

The model is optimized using a weighted loss that combines L1 loss, SSIM loss, and VGG perceptual loss:

$$L_{\text{total}} = 0.5L_1 + 0.3(1 - \text{SSIM}) + 0.2L_{\text{VGG}}. \quad (1)$$

The L1 term measures absolute pixel differences between the restored image and the clean target. The SSIM term encourages structural similarity. The VGG perceptual term compares higher-level feature representations and encourages visually meaningful details. The weights in (1) are the weights used in the project.

1) L1 Loss: L1 loss penalizes the absolute difference between corresponding pixels. It provides a direct pixel-level reconstruction objective and is defined as

$$L_1 = \frac{1}{N} \sum_{i=1}^{N} |I_i - \hat{I}_i|, \quad (2)$$

where I denotes the clean target, $\hat{I}$ denotes the predicted image, and N is the number of pixel values.

2) SSIM Loss: SSIM measures similarity in luminance, contrast, and local structure. It is computed with the `pytorch-msssim` library on images in the range $[0, 1]$. In the combined objective, the term $(1 - \text{SSIM})$ is minimized. This encourages the output to retain structural patterns that are important for the shape of handwritten characters.

3) VGG Perceptual Loss: The VGG perceptual component compares feature representations of the restored image and the clean target. A pretrained VGG16 network, truncated after its third convolutional block and kept frozen, extracts the features, and the L1 distance between the two feature maps is used as L_{VGG}. Instead of considering only individual pixel differences, it provides a feature-level comparison. It contributes 0.2 of the weighted total objective.

V. Training Process

Training is performed in Google Colab using an NVIDIA T4 GPU. The configuration of the final model is summarized in Table II. The combined dataset is trained for 12 epochs with a batch size of 4. Adam is used as the optimizer with an initial learning rate of 2×10^{-4}, and a ReduceLROnPlateau scheduler halves the learning rate when the validation loss stops improving.

A. Training and Validation

The training loop uses blurred images as inputs and clean images as targets. For each batch, the predicted output is passed to the combined loss function in (1). Gradients are calculated through backpropagation and the optimizer updates the model parameters. After every epoch, the model is evaluated on the held-out 20% of the pairs, the scheduler is updated with the validation loss, and the weights with the lowest validation loss are saved as the best checkpoint.

Development stages: The final configuration was reached step by step. First, an L1-only baseline with the same U-Net was trained on the HCCD pairs at 256×256 resolution for 20

TABLE II
TRAINING CONFIGURATION OF THE FINAL MODEL

Parameter	Setting
Environment	Google Colab
GPU	NVIDIA T4
Input resolution	512×512
Batch size	4
Optimizer	Adam
Initial learning rate	2×10^{-4}
Epochs	12
Scheduler	ReduceLROnPlateau (factor 0.5, patience 3)
Loss	$0.5L_1 + 0.3(1 - \text{SSIM}) + 0.2L_{\text{VGG}}$
Train/validation split	80:20 (random, fixed seed)
Model parameters	7.7 million

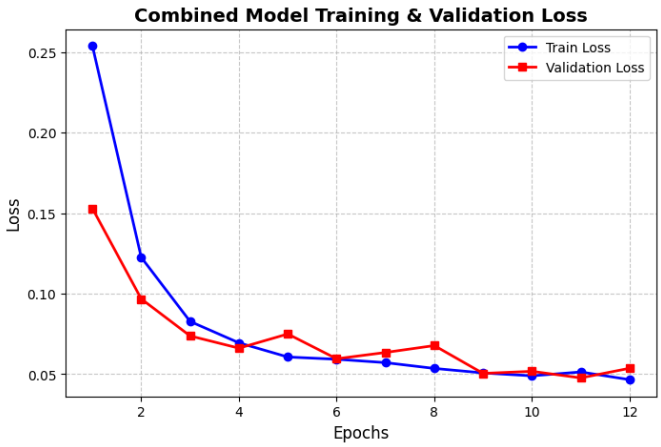


Fig. 2. Training and validation loss across the 12 epochs.

epochs (Adam, learning rate 10^{-3}, batch size 8). Second, an L1 + SSIM objective (weights 0.8 and 0.2) was tested at the same resolution. Third, the VGG perceptual term was added and the resolution was raised to 512×512. Finally, the Bangla pairs were added and the model was trained on the combined collection for 12 epochs.

Fig. 2 shows the training and validation loss curves of the final model. The plotted losses decrease overall during the 12 epochs, although small fluctuations are visible. The best validation loss is 0.0477.

VI. System Architecture and Workflow

The system is organized into six sequential stages: dataset input, data preprocessing, the deep learning model, loss calculation and backpropagation, inference and post-processing, and the Gradio user interface. Fig. 3 presents the detailed workflow diagram of the complete architecture.

A. Stage 1–6 Explanation

1) Stage 1: Dataset Input: The system receives paired English and Bangla handwriting data. English examples contain the HCCD blur levels 1, 2, and 3. Bangla examples are generated by applying synthetic Gaussian blur to clear handwriting images. Each blurred input is associated with a clean target.

2) Stage 2: Data Preprocessing: The input and target images are resized to 512×512 and converted to tensors with values in $[0, 1]$. This provides a consistent representation for model training and inference.

3) Stage 3: Deep Learning Model: The blurred input is processed by the U-Net, which contains an encoder, a bottleneck, and a decoder. Encoder features are connected to corresponding decoder stages through skip connections, and the restored image is produced through the sigmoid output layer.

4) Stage 4: Loss Calculation and Backpropagation: The predicted image and clean target are used to calculate the weighted L1, SSIM, and VGG perceptual objective in (1). Backpropagation computes gradients, and Adam updates the model parameters.

5) Stage 5: Inference and Post-processing: During inference, the trained model receives a user-provided blurry image, which is resized to 512×512 and passed through the network. The restored output is resized back to the original dimensions with Lanczos resampling, and a mild sharpness (factor 1.8) and contrast (factor 1.1) enhancement is applied. A patch-based (tiled) inference variant was also tested, but the final interface uses the single-pass method.

6) Stage 6: Gradio User Interface: A Gradio web interface allows a user to upload a handwritten image and view the deblurred output. This interface provides a practical way to test the trained model without directly running the training notebook.

VII. Evaluation Plan and Results

The final model is assessed using a 20% validation split and visual inspection, while PSNR and SSIM are reported for the L1 baseline. The visual examples include English handwriting at HCCD blur levels 1, 2, and 3 (Figs. 4–6), along with a Bangla handwriting example (Fig. 7). Each screenshot displays an input image and its restored output. The visual inspection also notes recovery of red ink traces, graph-paper lines, and distinct Bangla characters. Some examples are taken from the dataset and may overlap with the training data, so these screenshots are qualitative illustrations and do not replace a quantitative evaluation on a separate test set.

A. English Restoration: Level 1

Fig. 4 shows the restoration result for HCCD blur level 1.

B. English Restoration: Level 2

Fig. 5 shows the restoration result for HCCD blur level 2.

C. English Restoration: Level 3

Fig. 6 shows the restoration result for HCCD blur level 3.

D. Bangla Restoration

Fig. 7 shows a restoration example for Bangla handwriting, where the blurred input was generated with synthetic Gaussian blur.

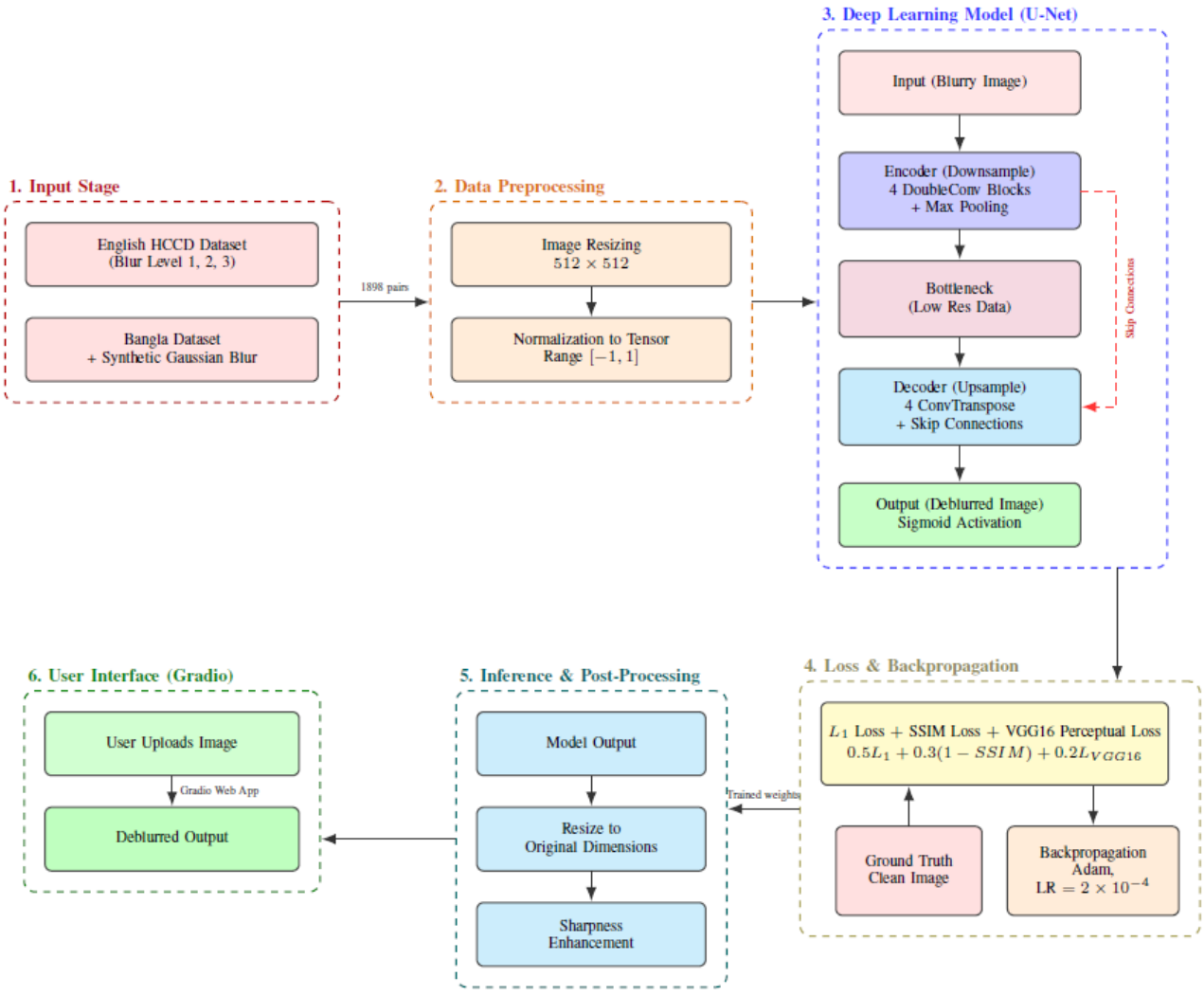


Fig. 3. Detailed workflow diagram of the proposed system.

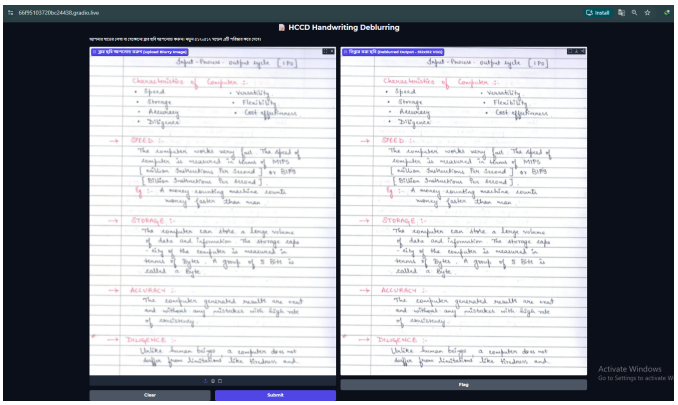


Fig. 4. English handwriting restoration example (Level 1).

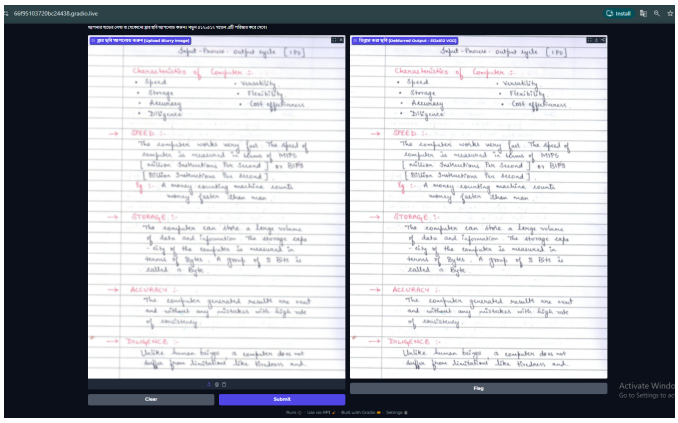


Fig. 5. English handwriting restoration example (Level 2).

E. Quantitative Metrics

Table III lists the reported metric values. PSNR and SSIM were computed with scikit-image on the validation split of the HCCD pairs (all three blur levels) for the L1-only baseline described in Section V. The best validation loss belongs to the final combined model and is the value of the weighted

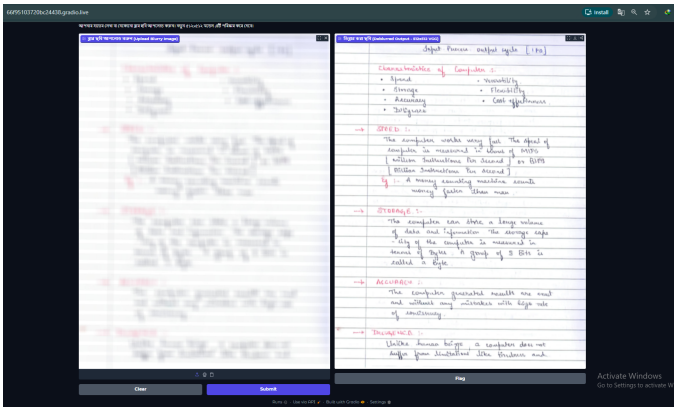


Fig. 6. English handwriting restoration example (Level 3).

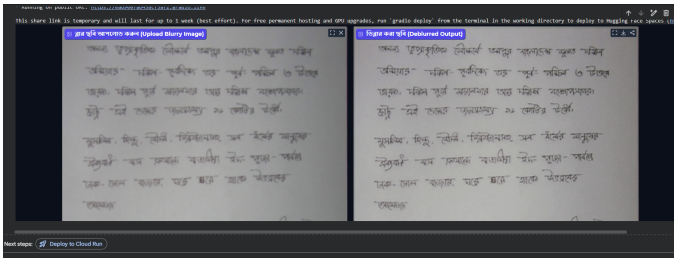


Fig. 7. Bangla handwriting restoration example.

TABLE III
REPORTED QUANTITATIVE RESULTS

Metric	Value	Model and data
PSNR	35.89 dB	L1 baseline, HCCD, 256 × 256
SSIM	> 0.96	L1 baseline, HCCD, 256 × 256
Best validation loss	0.0477	Final model, HCCD + Bangla, 512 × 512

objective in (1), not a plain L1 error. The two groups of values therefore come from different models and should not be compared directly.

The source experiments do not provide PSNR and SSIM for the final combined model, nor a per-language test table. These values should therefore be interpreted within the scope of the reported experiments.

VIII. Contribution to Sustainable Development Goals

The project relates to two United Nations Sustainable Development Goals (SDGs). First, SDG 4 (Quality Education) is relevant because restoration can support the digitization and readability of educational notes and handwritten learning materials. Second, SDG 9 (Industry, Innovation, and Infrastructure) is relevant because the work applies a deep learning method and a lightweight web interface to a document-processing problem. For low-resource languages such as Bangla, improving the legibility of handwritten documents may also support future OCR workflows.

IX. Limitations, Bias, Ethics, and Development Challenges

A. Limitations

The model may struggle with extreme real-world motion blur and heavily noisy backgrounds. The Bangla portion is generated using synthetic Gaussian blur with a kernel size of 15, which does not necessarily represent the motion blur, defocus, uneven illumination, or camera artifacts found in real photographs. PSNR and SSIM were not measured for the final combined model. In addition, the 80:20 split was made at the level of individual pairs, so the same page may appear at different blur levels in both the training and validation subsets; the validation loss may therefore be optimistic. Finally, resizing every image to a square 512 × 512 input distorts the aspect ratio of document pages.

B. Dataset Bias

The HCCD portion mainly contains English handwriting on graph paper, whereas the Bangla collection contains different handwriting styles. The combined dataset is also imbalanced: the English portion contributes 1,722 pairs and the Bangla portion contributes 176 pairs. These differences may lead to variation in performance across languages, writers, paper types, and writing styles.

C. Ethical Considerations

Handwriting can contain personal information. The datasets used in this work are not known to expose sensitive personal data. For future data collection and deployment, consent, access control, and careful handling of handwritten documents should be considered, particularly when documents contain names, addresses, academic records, or other personal details.

D. Development Challenges

The reported development challenges include Google Colab session disconnections, Google Drive mounting problems, GPU memory limitations, and out-of-memory errors when attempting higher image resolutions such as 1024 × 1024. The lack of naturally blurred paired Bangla data is another important challenge because it limits the diversity of degradation represented during training.

X. AI Usage Declaration

Generative AI tools were used for code debugging, for generating the synthetic Gaussian blur code, for assisting with parts of the training and interface code, and for structuring and editing this report. The authors ran the experiments, selected the configurations, and checked the results reported here.

XI. Conclusion and Future Work

This project presents a U-Net-based approach for restoring English and Bangla handwritten images. The system combines HCCD examples with synthetically blurred Bangla handwriting and trains a U-Net using L1, SSIM, and VGG perceptual losses. The best validation loss of the final model is 0.0477, an L1 baseline reaches about 35.89 dB PSNR on the HCCD validation

split, and the project includes a Gradio interface for image upload and restoration. The qualitative examples demonstrate the intended workflow for both languages.

The current approach has limitations, particularly the use of synthetic Gaussian blur for Bangla, the missing quantitative evaluation of the final model, and the difficulty of handling severe real-world motion blur and noisy backgrounds. Future work includes:

1) computing PSNR and SSIM for the final model on a page-disjoint test set, reported separately for English and Bangla;
2) training at 1024×1024 resolution or with aspect-ratio-preserving patches to investigate higher-definition output, subject to available GPU memory;
3) collecting real-world blurry Bangla documents to improve the range of training degradations; and
4) deploying the Gradio application to a permanent cloud service, such as Hugging Face Spaces.

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